

Tramacastilla_disturbance_palaeoecology_figures

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This is the code used to obtain figures and tables included in Julián-Posada et al. (in prep).

1. Table 1 - Results summary

Pollen data

```
pollen_summary <- data.frame(  
  Proxy = "Pollen",  
  Metric = "Number of pollen grains per sample",  
  Min = min(tram20_pollen_holocene$`Terrestrial pollen sum`),  
  Max = max(tram20_pollen_holocene$`Terrestrial pollen sum`),  
  Mean = round(mean(tram20_pollen_holocene$`Terrestrial pollen sum`), 2),  
  SD = round(sd(tram20_pollen_holocene$`Terrestrial pollen sum`), 2)) %>%  
  rbind(data.frame(  
    Proxy = "Pollen",  
    Metric = "Pollen richness: number of pollen types per sample",  
    Min = min(tram20_pollen_holocene$`No of taxa`),  
    Max = max(tram20_pollen_holocene$`No of taxa`),  
    Mean = round(mean(tram20_pollen_holocene$`No of taxa`), 2),  
    SD = round(sd(tram20_pollen_holocene$`No of taxa`), 2))) %>%  
  rbind(data.frame(  
    Proxy = "Pollen",  
    Metric = "PAR (number of grains·cm-2·yr-1)",  
    Min = round(min(tram20_PAR_sum$sum), 2),  
    Max = round(max(tram20_PAR_sum$sum), 2),  
    Mean = round(mean(tram20_PAR_sum$sum), 2),  
    SD = round(sd(tram20_PAR_sum$sum), 2)))
```

Fungal spores' data

```
fungal_spores_summary <- data.frame(  
  Proxy = "Fungal spores'",  
  Metric = "Number of fungal spores per sample",  
  Min = min(tram20_spores$total_spores),  
  Max = max(tram20_spores$total_spores),  
  Mean = round(mean(tram20_spores$total_spores), 2),  
  SD = round(sd(tram20_spores$total_spores), 2)) %>%  
  rbind(data.frame(  
    Proxy = "Fungal spores'",
```

```

Metric = "Fungal richness: number of fungal types per sample",
Min = min(tram20_spores$Noftaxa),
Max = max(tram20_spores$Noftaxa),
Mean = round(mean(tram20_spores$Noftaxa), 2),
SD = round(sd(tram20_spores$Noftaxa), 2)) %>%
rbind(data.frame(
  Proxy = "Fungal spores",
  Metric = "SAR (number of spores·cm-2·yr-1)",
  Min = round(min(tram20_SAR$total_SAR), 2),
  Max = round(max(tram20_SAR$total_SAR), 2),
  Mean = round(mean(tram20_SAR$total_SAR), 2),
  SD = round(sd(tram20_SAR$total_SAR), 2)))

```

Charcoal data

```

charcoal_summary <- data.frame(
  Proxy = "Charcoal",
  Metric = "Number of charcoal particles per sample",
  Min = min(CharTotS.110.age$CharCount),
  Max = max(CharTotS.110.age$CharCount),
  Mean = round(mean(CharTotS.110.age$CharCount), 2),
  SD = round(sd(CharTotS.110.age$CharCount), 2)) %>%
rbind(data.frame(
  Proxy = "Charcoal",
  Metric = "Charcoal area per sample (mm2)",
  Min = min(CHAR$sumCharArea)/1e6,
  Max = round(max(CHAR$sumCharArea)/1e6, 2),
  Mean = round(mean(CHAR$sumCharArea/1e6), 2),
  SD = round(sd(CHAR$sumCharArea/1e6), 2)) %>%
rbind (data.frame(
  Proxy = "Charcoal",
  Metric = "Width to length (W/L) ratio",
  Min = round(min(seedles$NPath.WLRatio), 2),
  Max = round(max(seedles$NPath.WLRatio), 2),
  Mean = round(mean(seedles$NPath.WLRatio), 2),
  SD = round(sd(seedles$NPath.WLRatio), 2)) %>%
rbind (data.frame(
  Proxy = "Charcoal",
  Metric = "CHAR (µm2·cm-2·yr-1)",
  Min = min(CHAR$CHAR.tram),
  Max = max(CHAR$CHAR.tram),
  Mean = round(mean(CHAR$CHAR.tram), 2),
  SD = round(sd(CHAR$CHAR.tram), 2)))

```

SedaDNA data

```

RAI_plants_sum <- RAI_plants %>%
  mutate(total_RAI_plants = rowSums(across(`Abies alba`:Ziziphora))) %>%
  mutate(richness_plants = rowSums(select(., `Abies alba`:Ziziphora) > 0, na.rm = TRUE)) %>%
  select(total_RAI_plants, richness_plants)

RAI_animals_sum <- RAI_animals %>%
  mutate(total_RAI_animals = rowSums(across(`Bos taurus`:Nyctalus noctula`))) %>%

```

```

mutate(richness_animals = rowSums(select(., `Bos taurus`:`Nyctalus noctula`) > 0, na.rm = TRUE)) %>%
select(total_RAI_animals, richness_animals)

sedaDNA_summary <- data.frame(
  Proxy = "sedaDNA",
  Metric = "Plant taxa per sample",
  Min = min(RAI_plants_sum$richness_plants),
  Max = max(RAI_plants_sum$richness_plants),
  Mean = round(mean(RAI_plants_sum$richness_plants), 2),
  SD = round(sd(RAI_plants_sum$richness_plants), 2)) %>%
  rbind(data.frame(
    Proxy = "sedaDNA",
    Metric = "Animal taxa per sample",
    Min = min(RAI_animals_sum$richness_animals),
    Max = max(RAI_animals_sum$richness_animals),
    Mean = round(mean(RAI_animals_sum$richness_animals), 2),
    SD = round(sd(RAI_animals_sum$richness_animals), 2))) %>%
  rbind(data.frame(
    Proxy = "sedaDNA",
    Metric = "Plant RAI",
    Min = round(min(RAI_plants_sum$total_RAI_plants), 2),
    Max = round(max(RAI_plants_sum$total_RAI_plants), 2),
    Mean = round(mean(RAI_plants_sum$total_RAI_plants), 2),
    SD = round(sd(RAI_plants_sum$total_RAI_plants), 2))) %>%
  rbind(data.frame(
    Proxy = "sedaDNA",
    Metric = "Animal RAI",
    Min = round(min(RAI_animals_sum$total_RAI_animals), 2),
    Max = round(max(RAI_animals_sum$total_RAI_animals), 2),
    Mean = round(mean(RAI_animals_sum$total_RAI_animals), 2),
    SD = round(sd(RAI_animals_sum$total_RAI_animals), 2)))

```

Create table 1

```

table1 <- rbind(pollen_summary, fungal_spores_summary, charcoal_summary, sedaDNA_summary)
table1

```

##	Proxy	Metric	Min
## 1	Pollen	Number of pollen grains per sample	313.00
## 2	Pollen	Pollen richness: number of pollen types per sample	14.00
## 3	Pollen	PAR (number of grains·cm ⁻² ·yr ⁻¹)	307.38
## 4	Fungal spores'	Number of fungal spores per sample	0.00
## 5	Fungal spores'	Fungal richness: number of fungal types per sample	0.00
## 6	Fungal spores'	SAR (number of spores·cm ⁻² ·yr ⁻¹)	0.00
## 7	Charcoal	Number of charcoal particles per sample	0.00
## 8	Charcoal	Charcoal area per sample (mm ²)	0.00
## 9	Charcoal	Width to length (W/L) ratio	0.05
## 10	Charcoal	CHAR (µm ² ·cm ⁻² ·yr ⁻¹)	0.00
## 11	sedaDNA	Plant taxa per sample	12.00
## 12	sedaDNA	Animal taxa per sample	0.00
## 13	sedaDNA	Plant RAI	0.90
## 14	sedaDNA	Animal RAI	0.00
##	Max	Mean	SD

```
## 1    604.00  432.78  61.74
## 2     47.00   29.63   7.41
## 3  13867.52 3955.97 2865.67
## 4     41.00   10.05   9.16
## 5     13.00    4.97   3.19
## 6    219.90   64.93  53.20
## 7     83.00    6.33   7.35
## 8      4.23    0.23   0.37
## 9      1.00    0.55   0.26
## 10 32926.81 2298.20 3017.64
## 11    208.00  112.41  43.44
## 12     10.00    3.09   3.07
## 13      1.00    0.98   0.02
## 14      1.00    0.57   0.42
```

2. Figure 2 - sedimentation rate + spores + animals

Include all data in one dataframe

```
sed.rate.SAR.diagram <- tram20_SAR %>%
  dplyr::select(-c(depth, total_SAR)) %>%
  left_join(tram20_depth_ages_sed_rate %>%
    dplyr::select(age, sed_rate), by = "age") %>%
  relocate(sed_rate, .after = "age") %>%
  relocate(`Obligate coprophilous`, .after = "Facultative coprophilous") %>%
  full_join(RAI_animals_domestic, by = "age") %>%
  arrange(age) %>%
  filter(age < 12000) %>%
  mutate(sed_rate = sed_rate * 1000, # CAREFUL WITH THIS TRANSFORMATION -- just for visualisation
         Cattle = Cattle * 100, # CAREFUL WITH THIS TRANSFORMATION -- just for visualisation
         Sheep = Sheep * 100, # CAREFUL WITH THIS TRANSFORMATION -- just for visualisation
         Goat = Goat * 100) # CAREFUL WITH THIS TRANSFORMATION -- just for visualisation
```

Configuration of styles

```
ex.spores <- c(FALSE, rep(TRUE, times = 28), rep(FALSE, times = 3))
c.sed.rate.line <- c("black", rep(NA, times = 31))
c.spores.bar <- c(NA, rep("black", times = 28), NA, NA, NA)
c.animals.bar <- c(rep(NA, times = 29), rep("#E57E7E", times = 3))
c.spores.poly <- c(NA, rep("#458B74", times = 14), rep("#993F00", times = 4),
  rep("orange", times = 8), rep("#458B74", times = 1), rep("#993F00", times = 1), NA, NA)
```

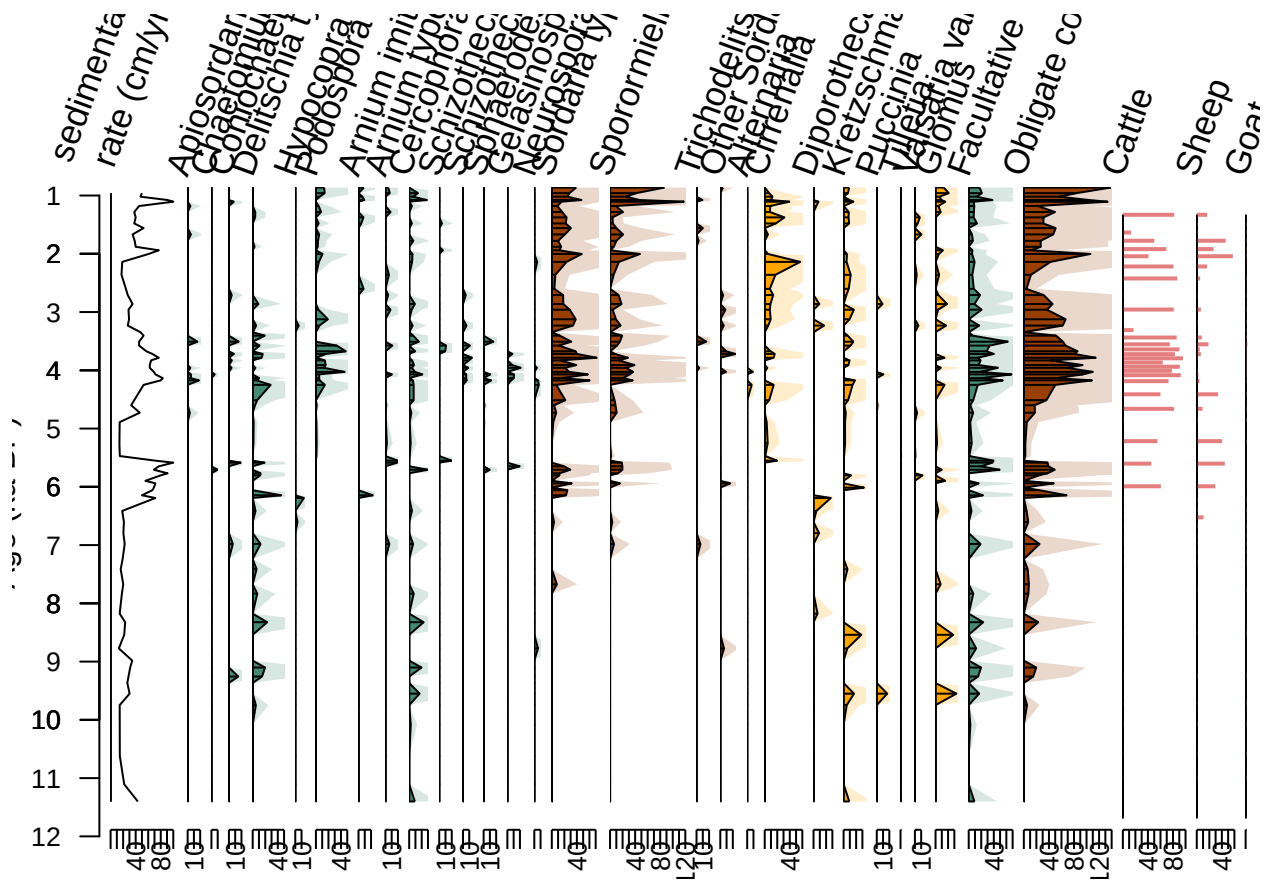
Create figure

```
p.sedrate.sar <- strat.plot(sed.rate.SAR.diagram %>%
  dplyr::select(-age) %>%
  rename("sedimentation \n rate (cm/yrs)" = "sed_rate"),
  yvar = sed.rate.SAR.diagram %>%
  dplyr::select(age) / 1000,
  ylabel = "Age (ka BP)", y.tks = seq(1, 12, 1), y.rev = TRUE,
  plot.line = TRUE, plot.poly = FALSE, plot.bar = FALSE, col.line = c.sed.rate.line,
```

```

scale.percent=TRUE, xSpace=0.009, x.pc.inc=10, x.pc.lab=TRUE, x.pc.omit0=TRUE)
p.sedrate.sar <- strat.plot(sed.rate.SAR.diagram %>%
  dplyr::select(-age),
  yvar = sed.rate.SAR.diagram %>%
  dplyr::select(age)/1000, y.rev=TRUE,
  plot.line=TRUE, plot.poly=TRUE, plot.bar=TRUE, col.line=c.spores.bar, col.bar=c.spores.bar,
  scale.percent=TRUE, xSpace=0.009, x.pc.inc=10, x.pc.lab=FALSE, x.pc.omit0=TRUE)
p.sedrate.sar <- strat.plot(sed.rate.SAR.diagram %>%
  dplyr::select(-age),
  yvar = sed.rate.SAR.diagram %>%
  dplyr::select(age)/1000, y.rev=TRUE,
  plot.line=FALSE, plot.poly=FALSE, plot.bar=TRUE, col.bar= c.animals.bar, lw=1,
  scale.percent=TRUE, xSpace=0.009, x.pc.inc=10, x.pc.lab=FALSE, x.pc.omit0=TRUE)
x.axis=FALSE, x.names = NA, y.pc.lab = FALSE, add=TRUE) #save pdf A4 and MO

```



3. Figure 3 - Changes in disturbance regimes with RoC, cpt and fire events

Figure 3a - Fire events

```

p.fig3.1 <- tr_glob_exp %>%
  ggplot(aes(x = age_top_i/1000, y = acc_i)) +

```

```

geom_col(width = 0.01, fill = "grey40") +
geom_smooth(method = "loess", span = 0.15, se = FALSE, color = "red", size = 0.5) +
geom_smooth(method = "loess", span = 0.2, se = FALSE, color = "grey", size = 0.5) +
geom_point(data = tr_glob_exp %>% filter(peaks_pos_sig > 0),
           aes(x = age_top_i/1000, y = peaks_pos_sig*30000),
           shape = 3, size = 2, color = "red") +
geom_col(data = tr_glob_exp %>% filter(peaks_pos_sig > 0),
          aes(x = age_top_i/1000, y = peaks_pos_sig*30000),
          fill = "red", alpha = 0.3, width = 0.02) +
scale_x_reverse(limits = c(12, 0.5),
                breaks = seq(0.5, 12, by = 0.5),
                labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
                                "")) +
scale_y_continuous(breaks = c(0, 10000, 20000),
                   labels = c("0", "1e+4", "2e+4")) +
theme_classic() +
ylab("") +
theme(axis.title.x = element_blank(),
      legend.position = "right",
      plot.margin = margin(0, 0, 0, 0, "cm"),
      legend.text = element_text(size = 11))

```

Figure 3b - W/L ratio

```

p.fig3.2 <- W.L.ratio %>%
  rename("Type of fuel" = "fuel") %>%
  ggplot(aes(x = age/1000, y = count, fill = `Type of fuel`, color = `Type of fuel`)) +
  geom_area(alpha = 0.6, position = "stack") +
  geom_smooth(method = "loess", se = FALSE, aes(color = `Type of fuel`), linewidth = 1, alpha = 0.8) +
  scale_y_continuous(trans = "sqrt", # transformation with squared root
                    breaks = c(0, 5, 20, 50)) +
  labs(y = "") +
  theme_classic() +
  scale_x_reverse(limits = c(12, 0.5),
                  breaks = seq(0.5, 12, by = 0.5),
                  labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
                                  "")) +
  scale_fill_manual(values = c("Woody" = "darkgreen", "Grasslands" = "#FFD700")) +
  scale_color_manual(values = c("Woody" = "darkgreen", "Grasslands" = "#FFD700")) +
  theme(axis.title.x = element_blank(),
        legend.position = "right",
        plot.margin = margin(0, 0, 0, 0, "cm"),
        legend.text = element_text(size = 11)) +
  annotate("text", x = 2, y = 120, label = "peak = 59", vjust = -1, color = "red")

```

Figure 3c - CHAR with changepoints

```

p.fig3.3 <- tram20_CHARcpt %>%
  ggplot(aes(x = age/1000, y = CHAR.tram)) +
  geom_line() +
  geom_vline(xintercept = changepoint_ages_CHAR/1000, color = "red", linetype = "dashed") +
  theme_classic() +
  scale_x_reverse(limits = c(12, 0.5),
                  breaks = seq(0.5, 12, by = 0.5),
                  labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
                                  "")) +

```

```

scale_y_continuous(breaks = c(0, 10000, 20000, 30000),
                   labels = c("0", "1e+4", "2e+4", "3e+4")) +
ylab("CHAR ( $\mu\text{m}^2\cdot\text{cm}^{-2}\cdot\text{yr}^{-1}$ ") +
theme(axis.title.x = element_blank())

```

Figure 3d - Rate of Change of SAR

```

p.fig3.4 <- scenario_4_SAR_peak %>%
ggplot(aes(x = Age/1000))+
theme_classic() +
geom_ribbon(aes(ymin = ROC_dw, ymax = ROC_up), fill = "grey") +
geom_line(aes(y = ROC)) +
geom_point(size = 3, aes(y = ROC_peak), shape = 21, fill = "green") +
scale_x_reverse(limits = c(12, 0.5),
                breaks = seq(0.5, 12, by = 0.5),
                labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
                                "")) +
ylab("RoC of SAR") +
theme(axis.title.x = element_blank())

```

Figure 3e - Rate of Change of PAR

```

p.fig3.5 <- scenario_4_PAR_peak %>%
ggplot(aes(x = Age/1000))+
theme_classic() +
geom_ribbon(aes(ymin = ROC_dw, ymax = ROC_up), fill = "grey") +
geom_line(aes(y = ROC)) +
geom_point(size = 3, aes(y = ROC_peak), shape = 21, fill = "green") +
scale_x_reverse(limits = c(12, 0.5),
                breaks = seq(0.5, 12, by = 0.5),
                labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
                                "")) +
labs(y = "RoC of PAR") +
theme(axis.title.x = element_blank())

```

Figure 3f - Rate of Change of RAI

```

p.fig3.6 <- scenario_4_RAI_peak %>%
ggplot(aes(x = Age/1000))+
theme_classic() +
geom_ribbon(aes(ymin = ROC_dw, ymax = ROC_up), fill = "grey") +
geom_line(aes(y = ROC)) +
geom_point(size = 3, aes(y = ROC_peak), shape = 21, fill = "green") +
scale_x_reverse(limits = c(12, 0.5),
                breaks = seq(0.5, 12, by = 0.5),
                labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
                                "")) +
labs(x = "Age (ka BP)", y = "RoC of RAI")

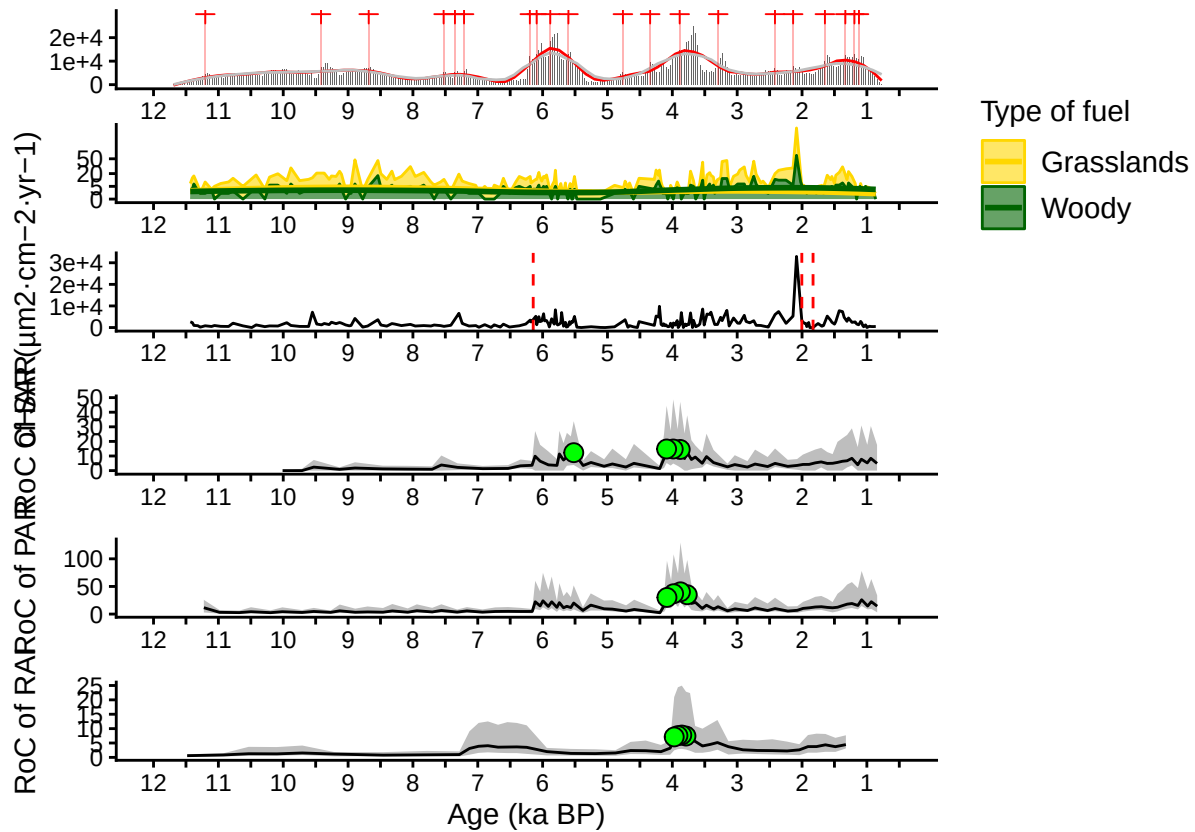
```

Create figure 3

```

p.fig3.1/p.fig3.2/p.fig3.3/p.fig3.4/p.fig3.5/p.fig3.6 #save A5

```



4. Table 2 - Generalized Additive Models (GAMs)

```
extract <- function(summary_gam, var_name, predictor_name) {
  coef <- summary_gam$p.table[2,]
  data.frame(Variable=var_name, Predictor=predictor_name,
    Estimate=coef[1], `Standard error`=coef[2], `t-value`=coef[3],
    `p-value`=ifelse(coef[4]<0.001,"< 0.001 **",paste(round(coef[4],3),"**")),
    Deviance=paste0(round(summary_gam$dev.expl*100),"%"))
}

rbind(
  extract(summary_gam1, "Rate of Change sedaDNA", "Temperature"),
  extract(summary_gam2, "Rate of Change sedaDNA", "Cattle"),
  extract(summary_gam3, "Rate of Change sedaDNA", "Sheep"),
  extract(summary_gam4, "Rate of Change sedaDNA", "Obligate coprophilous fungi"),
  extract(summary_gam5, "Rate of Change sedaDNA", "Fire"),
  extract(summary_gam1_PAR, "Rate of Change pollen", "Temperature"),
  extract(summary_gam2_PAR, "Rate of Change pollen", "Cattle"),
  extract(summary_gam3_PAR, "Rate of Change pollen", "Sheep"),
  extract(summary_gam4_PAR, "Rate of Change pollen", "Obligate coprophilous fungi"),
  extract(summary_gam5_PAR, "Rate of Change pollen", "Fire"))
```

##	Variable	Predictor	Estimate
----	----------	-----------	----------


```

## Estimate Rate of Change sedaDNA Temperature 6.373558e-01
## Estimate1 Rate of Change sedaDNA Cattle 5.041046e-01
## Estimate2 Rate of Change sedaDNA Sheep -9.757928e-02
## Estimate3 Rate of Change sedaDNA Obligate coprophilous fungi 1.118051e+00
## Estimate4 Rate of Change sedaDNA Fire 2.944874e-05
## Estimate5 Rate of Change pollen Temperature 6.114644e-01
## Estimate6 Rate of Change pollen Cattle 5.276179e-01
## Estimate7 Rate of Change pollen Sheep 4.299549e-02
## Estimate8 Rate of Change pollen Obligate coprophilous fungi 8.320422e-01
## Estimate9 Rate of Change pollen Fire 6.911124e-06
## Standard.error t.value p.value Deviance
## Estimate 1.254662e-01 5.0798992 < 0.001 ** 41%
## Estimate1 1.417704e-01 3.5557806 0.001 ** 25%
## Estimate2 1.634371e-01 -0.5970450 0.554 ** 1%
## Estimate3 1.946514e-01 5.7438637 < 0.001 ** 47%
## Estimate4 6.731188e-05 0.4374969 0.664 ** 1%
## Estimate5 1.087930e-01 5.6204396 < 0.001 ** 46%
## Estimate6 1.209108e-01 4.3636975 < 0.001 ** 34%
## Estimate7 1.486435e-01 0.2892523 0.774 ** 0%
## Estimate8 2.003558e-01 4.1528228 < 0.001 ** 32%
## Estimate9 6.114221e-05 0.1130336 0.911 ** 0%

```

5. Table 3 - Parameters of Structural Equation Model (SEM)

```

parameterEstimates(fit_no_normal_8, standardized = TRUE) %>%
  filter(op == "~") %>%
  dplyr::select(lhs, rhs, est, se, z, std.all, pvalue) %>%
  rename("Response" = "lhs",
         "Predictor" = "rhs",
         "Estimate" = "est",
         "Standard error" = "se",
         "z-value" = "z",
         "Standardised estimate" = "std.all",
         "p-value" = "pvalue")

```

```

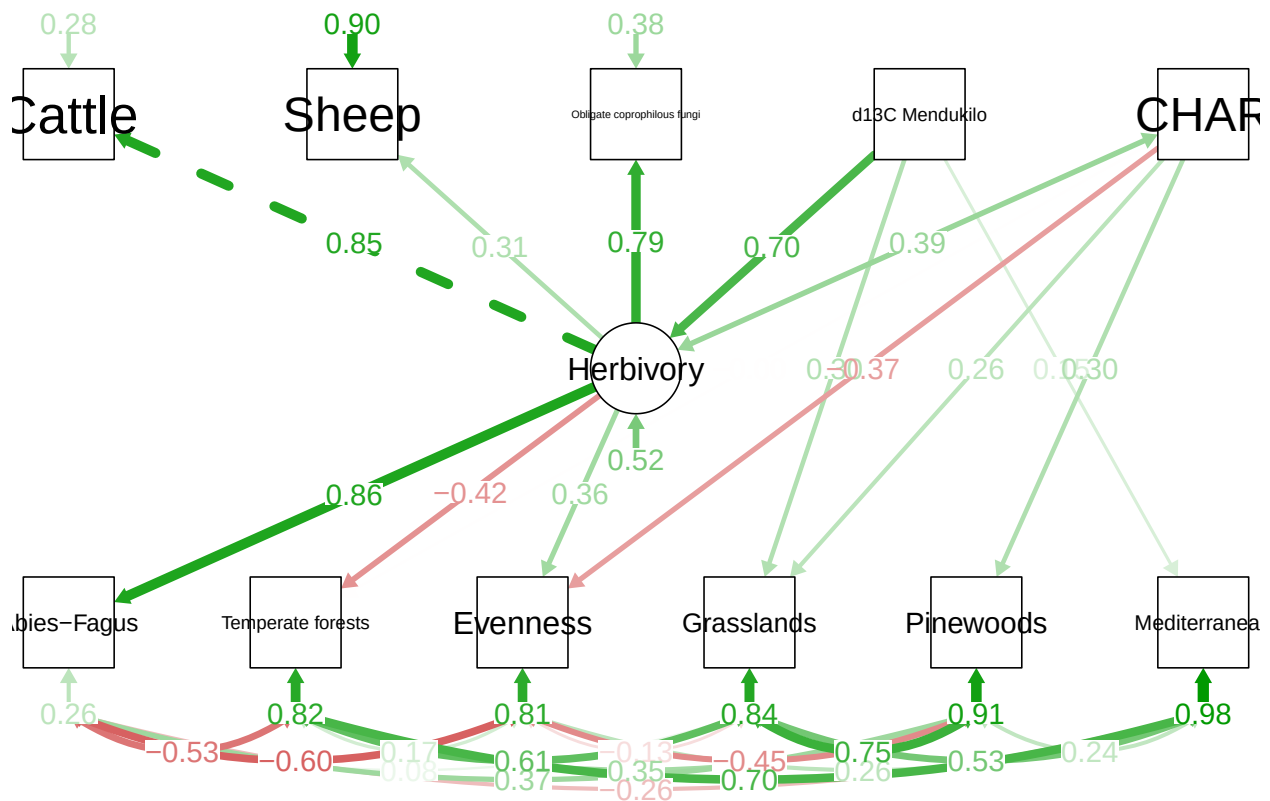
## Response Predictor Estimate Standard.error z.value
## 1 Herbivory z_d13C 0.606 0.113 5.381
## 2 log_PAR_Abies_Fagus Herbivory 1.227 0.172 7.129
## 3 log_PAR_grassland log_CHAR 0.123 0.056 2.217
## 4 log_PAR_grassland z_d13C 0.095 0.026 3.640
## 5 log_PAR_pinewoods log_CHAR 0.166 0.074 2.257
## 6 log_PAR_temperate log_CHAR -0.001 0.062 -0.023
## 7 log_PAR_temperate Herbivory -0.223 0.069 -3.231
## 8 log_PAR_mediterranean z_d13C 0.073 0.059 1.231
## 9 z_evenness_polen Herbivory 0.417 0.162 2.572
## 10 z_evenness_polen log_CHAR -0.558 0.204 -2.731
## Standardised.estimate p.value
## 1 0.696 0.000
## 2 0.863 0.000
## 3 0.261 0.027
## 4 0.299 0.000

```

```
## 5          0.303  0.024
## 6         -0.002  0.981
## 7         -0.419  0.001
## 8          0.150  0.218
## 9          0.359  0.010
## 10        -0.371  0.006
```

6. Figure 4 - Structural Equation Model (SEM)

```
semPaths(fit_no_normal_8,
  what = "std",
  whatLabels = "std",
  style = "lisrel",
  layout = "tree",
  nodeLabels = c("Cattle", "Sheep", "Obligate coprophilous fungi", "Abies-Fagus",
    "Temperate forests", "Evenness", "Grasslands", "Pinewoods", "Mediterranean", "d13C Mendukilo",
    "Herbivory"),
  edge.label.cex = 1.2,
  sizeMan = 8,
  sizeLat = 8,
  label.cex = 2,
  mar = c(8, 1, 6, 1))
```



7. Figure 5 - Cross-correlations

Cross correlations before 6.5 ka BP

```
fig_cross_before <- p.richness.ccf_before/p.evenness.ccf_before/p.poaceae.ccf_before/
p.pinus.ccf_before/p.betula.ccf_before/p.corylus.ccf_before/p.decquercus.ccf_before +
plot_layout(ncol = 1, nrow = 9) +
theme(plot.margin = margin(0, 0, 0, 0, "pt")) +
plot_annotation(title = "Before 6.5 ka BP")
```

Cross correlations after 6.5 ka BP

```
fig_cross_after <- p.richness.ccf_after/p.evenness.ccf_after/ p.poaceae.ccf_after/p.pinus.ccf_after/
p.betula.ccf_after/p.corylus.ccf_after/p.decquercus.ccf_after/p.abies.ccf_after/p.fagus.ccf_after +
plot_layout(ncol = 1, nrow = 9) +
theme(plot.margin = margin(0, 0, 0, 0, "pt")) +
plot_annotation(title = "After 6.5 ka BP")
```

Cross correlations before and after 6.5 ka BP

```
fig_cross_before|fig_cross_after +
plot_layout(guides = "collect")
```

